# CHAPTER 8 Alcohols, Phenols and Ethers

## Alcohols

**1.** Given below are two statements:

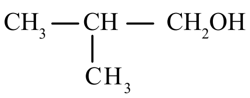
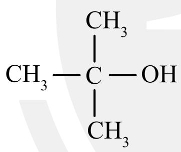
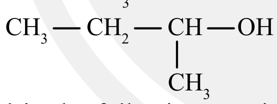
**Statement-I:** Propene on treatment with diborane gives an addition product with the formula $((CH_3)_2 - CH)_3B$.

**Statement-II:** Oxidation of $((CH_3)_2 - CH)_3B$ with hydrogen peroxide in presence of NaOH gives propan-2ol.

In the light of the above statements, choose the **most appropriate** answer from the options given below: *(2024 Re)*

a. Statement-I is correct but Statement-II is incorrect
b. Statement-I is incorrect but Statement-II is correct
c. Both Statement-I and Statement-II are correct
d. Both Statement-I and Statement-II are incorrect

**2.** Which one of the following alcohols reacts instantaneously with Lucas reagent? *(2024)*

a. 
b. 
c. $CH_3—CH_2—CH_2—CH_2O$
d. 

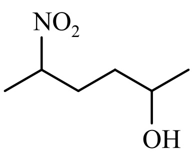
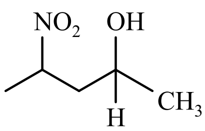
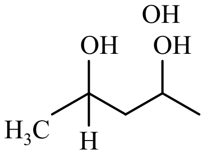
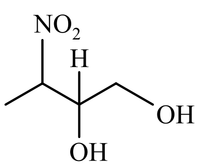
**3.** The products A and B obtained in the following reactions, respectively, are *(2024)*

$3ROH + PCl_3 \rightarrow 3RCl + A$

$ROH + PCl_5 \rightarrow RCl + HCl + B$

a. $H_3PO_4$ and $POCl_3$
b. $H_3PO_3$ and $POCl_3$
c. $POCl_3$ and $H_3PO_3$
d. $POCl_3$ and $H_3PO_4$

**4.** Which amongst the following will be most readily dehydrated under acidic conditions ? *(2023)*

a. 
b. 
c. 
d. 

**5.** $R–COOH \xrightarrow[(ii) H_2O/HCl]{(i) \text{“X”}} R–CH_2OH$

$R–CH = CH_2 \xrightarrow[(ii) H_2O, NaOH, H_2O_2]{(i) \text{“X”}} R–CH_2–CH_2–OH$

Identify ‘X’ in above reactions *(2023-Manipur)*

a. $B_2H_6$ b. $LiAlH_4$ c. $NaBH_4$ d. $H_2/Pd$

**6.** Reagents which can be used to convert alcohols to carboxylic acids, are *(2023-Manipur)*

A. $CrO_3–H_2SO_4$
B. $K_2Cr_2O_7 + H_2SO_4$
C. $KMnO_4 + KOH/H_3O^+$
D. Cu, 573 K
E. $CrO_3$, $(CH_3CO)_2O$

Choose the **most appropriate** answer form the options given below:

a. (B), (C) and (D) only
b. (B), (D) and (E) only
c. (A), (B) and (C) only
d. (A), (B) and (E) only

**7.** Given below are two statements: *(2022)*

**Statement-I:** In Lucas test, primary, secondary and tertiary alcohols are distinguished on the basis of their reactivity with conc. $HCl + ZnCl_2$, known as Lucas Reagent.

**Statement-II:** Primary alcohols are most reactive and immediately produce turbidity at room temperature on reaction with Lucas Reagent.

In the light of the above statements, choose the **most appropriate** answer from the options given below:

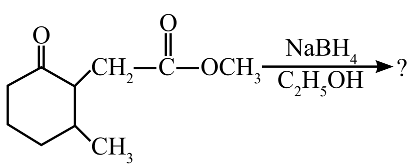
a. Statement-I is incorrect but Statement-II is correct.
b. Both Statement-I and Statement-II are correct.
c. Both Statement-I and Statement-II are incorrect.
d. Statement-I is correct but Statement-II is incorrect.

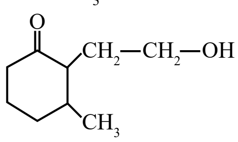
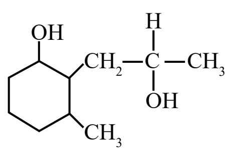
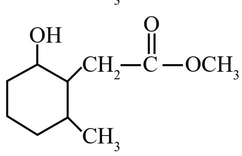
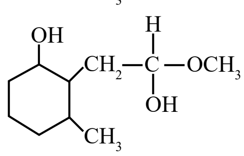
**8.** What is the IUPAC name of the organic compound formed in the following chemical reaction? *(2021)*

$$\text{Acetone} \xrightarrow[\text{(ii) } H_2O,\ H^+]{\text{(i) } C_2H_5MgBr,\ \text{dry ether}} \text{Product}$$

a. Pentan-2-ol
b. Pentan-3-ol
c. 2-methyl butan-2-ol
c. 2-methyl propan-2-ol

**9.** The product formed in the following chemical reaction is: *(2021)*



a. 
b. 
c. 
d. 

**10.** Reaction between acetone and methylmagnesium chloride followed by hydrolysis will give : *(2020)*

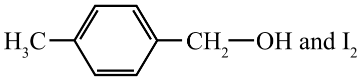
a. Sec. butyl alcohol  b. Tert. butyl alcohol
c. Isobutyl alcohol  d. Isopropyl alcohol

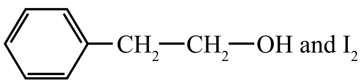
**11.** $CH_3CH_2CH=CH_2 \xrightarrow[H_2O,\ H_2O/OH]{B_2H_6} Z$

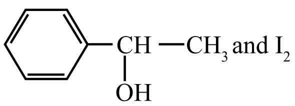
What is Z? *(2020-Covid)*

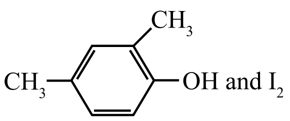
a. $CH_3CH_2CH(OH)CH_3$  b. $CH_3CH_2CH_2CHO$
c. $CH_3CH_2CH_2CH_3$  d. $CH_3CH_2CH_2CH_2OH$

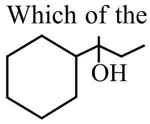
**12.** Compound A, $C_8H_{10}O$, is found to react with NaOI (produced by reacting Y with NaOH) and yields a yellow precipitate with characteristic smell. A and Y are respectively *(2018)*

a. 

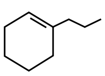
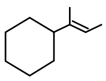
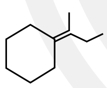
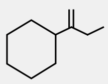

b. 


c. 


d. 


**13.** Which of the following is not the product of dehydration of 

? *(2015 Re)*

a.  b.  c.  d. 

## Phenols

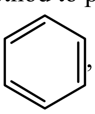
**14.** Match the reagents (List-I) with the product (List-II) obtained from phenol. *(2022 Re)*

| | List-I | | List-II |
|---|---|---|---|
| A. | (i) NaOH (ii) $CO_2$ (iii) $H^+$ | I. | Benzoquinone |
| B. | (i) Aqueous NaOH + $CHCl_3$ (ii) $H^+$ | II. | Benzene |
| C. | Zn dust, Δ | III. | Salicyl aldehyde |
| D. | $Na_2Cr_2O_7$, $H_2SO_4$ | IV. | Salicylic acid |

Choose the **correct answer** from the options given below:

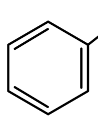
a. A-IV, B-II, C-I, D-III  b. A-III, B-IV, C-I, D-II
c. A-II, B-I, C-IV, D-III  d. A-IV, B-III, C-II, D-I

**15.** Which one of the following reaction sequence is **incorrect** method to prepare phenol? *(2022 Re)*

a. , oleum, NaOH, $H_3O^+$

b. Aniline, $NaNO_2$ + HCl, $H_2O$, heating

c. Cumene, $O_2$, $H_3O^+$

d. 

, NaOH, STP condition

**16.** Given below are two statements: *(2022)*

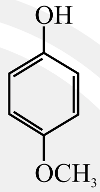
**Statement-I:** The acidic strength of monosubstituted nitrophenol is higher than phenol because of electron withdrowing nitro group.

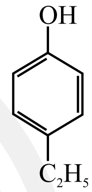
**Statement-II:** o-nitrophenol, m-nitrophenol and p-nitrophenol will have same acidic strength as they have one nitro group attached to the phenolic ring.

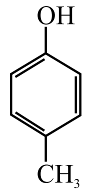
In the light of the above statements, choose the **most appropriate** answer form the options given below:

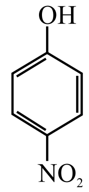
a. **Statement-I** is incorrect but **Statement-II** is correct.
b. Both **Statement-I** and **Statement-II** are correct.
c. Both **Statement-I** and **Statement-II** are incorrect.
d. **Statement-I** is correct but **Statement-II** is incorrect.

**17.** Which of the following substituted phenols is the strongest acid? *(2020-Covid)*

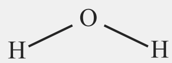
a. 

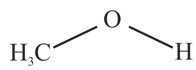

b. 

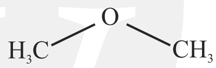

c. 

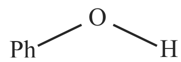

d. 


**18.** The compound that is most difficult to protonate is: *(2019)*

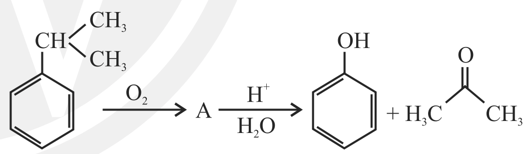
a. 

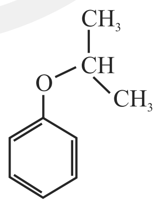

b. 

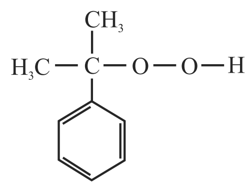

c. 

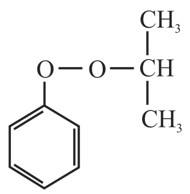

d. 

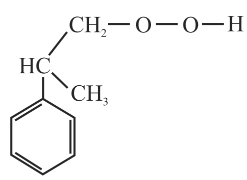

**19.** The structure of intermediate A in the following reaction, is *(2019)*



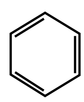

a. 


b. 


c. 

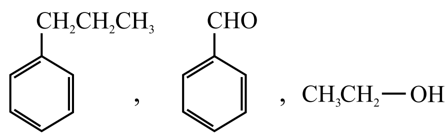

d. 

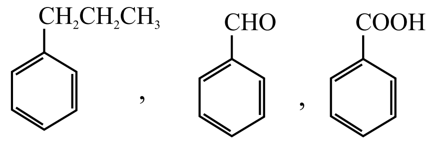

**20.** Identify the major products P, Q and R in the following sequence of reactions: *(2018)*

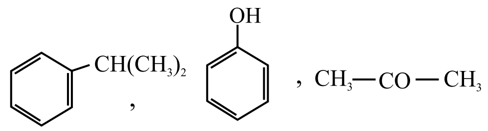
 + $CH_3CH_2CH_2Cl \xrightarrow[]{\text{Anhydrous } AlCl_3} P$

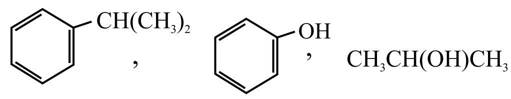
$$P \xrightarrow[\text{(ii) } H_2O/\Delta]{\text{(i)}O_2} Q + R$$

P Q R

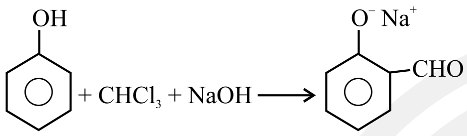
a. 


b. 


c. 


d. 


**21.** In the reaction, the electrophile involved is: *(2018)*



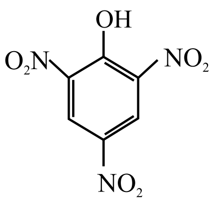

a. Dichloromethyl cation $\left(\overset{\oplus}{C}HCl_2\right)$

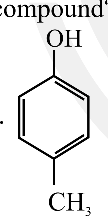
b. Formyl cation $\left(\overset{\oplus}{C}HO\right)$

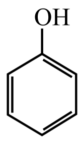
c. Dichlorocarbene $\left(:CCl_2\right)$

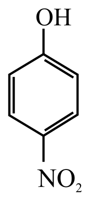
d. Dichloromethyl anion $\left(\overset{\oplus}{C}HCl_2\right)$

**22.** Which one is the most acidic compound? *(2017-Delhi)*

a. 


b. 


c. 


d. 


**23.** Reaction of phenol with chloroform in presence of dilute sodium hydroxide finally introduces which one of the following functional group? *(2015 Re)*

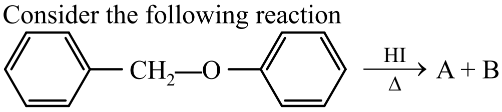
a. –CHO b. –$CH_2Cl$ c. –COOH d. –$CHCl_2$

**24.** Which of the following will not be soluble in sodium hydrogen carbonate? *(2014)*

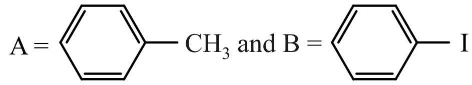
a. Benzoic acid b. o-Nitrophenol
c. Benzenesulphonic acid d. 2,4,6-trinitrophenol

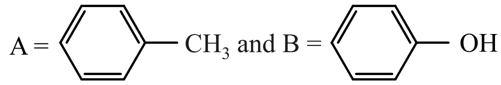
## Ethers

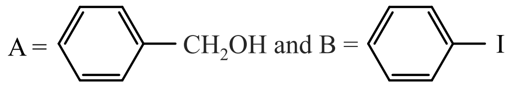
**25.** Consider the following reaction *(2023)*

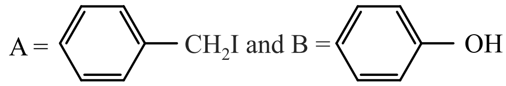



Identify products A and B

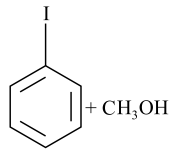
a. 

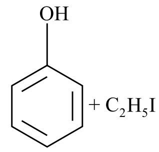

b. 

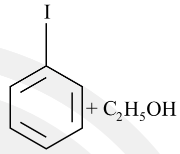

c. 

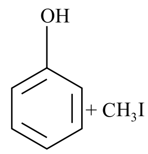

d. 


**26.** Anisole on cleavage with HI gives: *(2020)*

a. 


b. 


c. 


d. 

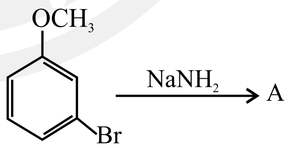

**27.** The compound A on treatment with Na gives B, and with $PCl_5$ gives C. B and C react together to give diethyl ether. A, B and C are in the order: *(2018)*

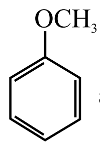
a. $C_2H_5OH$, $C_2H_6$, $C_2H_5Cl$
b. $C_2H_5OH$, $C_2H_5Cl$, $C_2H_5ONa$
c. $C_2H_5OH$, $C_2H_5ONa$, $C_2H_5Cl$
d. $C_2H_5Cl$, $C_2H_6$, $C_2H_5OH$

**28.** The heating of phenyl-methyl ethers with HI produces. *(2017-Delhi)*

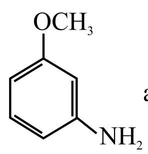
a. Benzene b. Ethyl chlorides
c. Iodobenzene d. Phenol

**29.** Identify A and predict the type of reaction: *(2017-Delhi)*

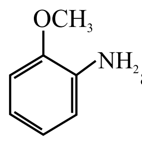



a. 

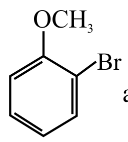
and cine substitution reaction

b. 

and substitution reaction

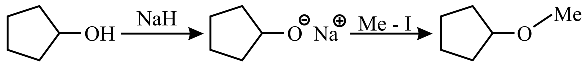
c. 

and elimination addition reaction

d. 

and cine substitution reaction

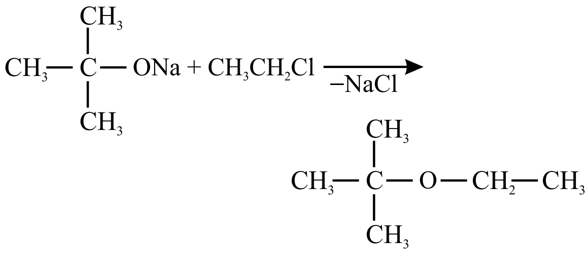
**30.** The reaction:




can be classified as? *(2016 - I)*

a. Williamson alcohol synthesis reaction
b. Williamson ether synthesis reaction
c. Alcohol formation reaction
d. Dehydration reaction

**31.** The reaction




is called: *(2015)*

a. Williamson continuous esterification process
b. Etard reaction
c. Gatterman - Koch reaction
d. Williamson synthesis

**32.** Among the following sets of reactants which one produces anisole? *(2014)*

a. $C_6H_5OH$; NaOH; $CH_3I$
b. $C_6H_5OH$; neutral $FeCl_3$
c. $C_6H_5 - CH_3$; $CH_3COCl$; $AlCl_3$
d. $CH_3CHO$; RMgX

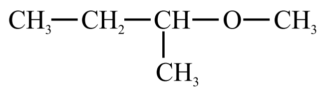
**33.** Identify Z in the sequence of reactions:

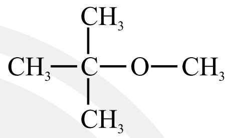
$$CH_3CH_2CH = CH_2 \xrightarrow{HBr/H_2O_2} Y \xrightarrow{C_2H_5ONa} Z \quad (2014)$$

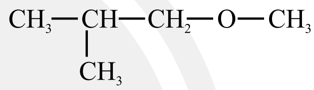
a. $(CH_3)_2CH - O - CH_2CH_3$
b. $CH_3(CH_2)_4 - O - CH_3$
c. $CH_3CH_2 - CH(CH_3) - O - CH_2CH_3$
d. $CH_3 - (CH_2)_3 - O - CH_2CH_3$

**34.** Among the following ethers, which one will produce methyl alcohol on treatment with hot concentrated HI? *(2013)*

a. $CH_3 - CH_2 - CH_2 - CH_2 - O - CH_3$

b. 


c. 


d. 


## Answer Key

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| **1.** (b) | **2.** (b) | **3.** (b) | **4.** (c) | **5.** (a) | **6.** (c) | **7.** (d) | **8.** (c) | **9.** (c) | **10.** (b) |
| **11.** (d) | **12.** (c) | **13.** (a) | **14.** (d) | **15.** (d) | **16.** (d) | **17.** (d) | **18.** (d) | **19.** (b) | **20.** (c) |
| **21.** (c) | **22.** (a) | **23.** (a) | **24.** (b) | **25.** (d) | **26.** (d) | **27.** (c) | **28.** (d) | **29.** (b) | **30.** (b) |
| **31.** (d) | **32.** (a) | **33.** (d) | **34.** (c) | | | | | | |